Entropy Collapse, Information Saturation, and Baby-Universe Genesis: A Detailed Mathematical Model

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Abstract

We present a comprehensive mathematical framework describing a pathway from black-hole evaporation and information dynamics to the formation of causally disconnected baby universes. The model synthesizes classical general relativity, black-hole thermodynamics, quantum-information (Page curve) constraints, a critical-density mass relation (the "Anas Limit"), an effective quantum-pressure correction to the Tolman–Oppenheimer–Volkoff (TOV) system, Israel-junction throat formation, and a decoherence-based interpretation of observation as disturbances in an information field. We derive explicit criteria for information-saturation-driven bounces, give order-of-magnitude numerical estimates for astrophysical parameters, and describe observational signatures. The manuscript is written as a single, continuous paper intended for further expansion into a full 50-page article with numerical relativity simulations and figures.

Contents

Introduction

This manuscript develops a hypothesis and a working mathematical model in which black holes, as they evaporate and approach strong quantum-gravitational regimes, can form an information-saturated interior core. When the information (entropy) density within the interior reaches a critical threshold determined by Planckian bounds and the chosen quantum-gravity mechanism, the interior experiences a bounce-like transition. The bounce produces a causally disconnected expanding region which can be interpreted as a nascent or "baby" universe connected transiently to the parent black hole via a throat (wormhole-like geometry).

This proposal synthesizes several existing strands of theoretical research:

- Black-hole thermodynamics (Bekenstein–Hawking entropy, Hawking radiation),
- Quantum information aspects of black holes (Page curve and unitary evolution),
- Singularity resolution ideas (loop quantum cosmology, quantum pressure terms),
- Wormhole / Einstein–Rosen bridge junction conditions,
- Decoherence theory and information-theoretic accounts of measurement.

The paper proceeds as a single continuous exposition: first establishing the thermodynamic and information-theoretic background, then deriving a critical-density mass criterion (the Anas Limit), introducing an effective TOV + quantum-pressure toy model, analyzing throat formation via junction conditions, deriving baby-universe FRW initial conditions, modeling decoherence/visibility, and finally proposing observational consequences and next steps.

1 Black-hole thermodynamics and information preliminaries

1.1 Bekenstein–Hawking entropy and Hawking temperature

For a non-rotating, uncharged Schwarzschild black hole of mass M the Schwarzschild radius r_s and horizon area A are

$$r_s = \frac{2GM}{c^2}, \quad A = 4\pi r_s^2 = 16\pi \frac{G^2M^2}{c^4}. \tag{1}$$

The Bekenstein–Hawking entropy S_{BH} and Hawking temperature T_H are

$$S_{BH} = \frac{k_B A}{4L_P^2} = \frac{4\pi k_B G^2 M^2}{\hbar c}, \tag{2}$$

$$T_H = \frac{\hbar c^3}{8\pi G k_B M}. \tag{3}$$

Here $L_P = \sqrt{\hbar G/c^3}$ is the Planck length and k_B Boltzmann's constant.

1.2 Evaporation timescale and mass loss rate

Hawking emission leads to mass loss at approximately

$$\frac{dM}{dt} = -\frac{\alpha}{M^2}, \qquad \alpha \approx \frac{\hbar c^4}{15360\pi G^2} \cdot \tilde{g}, \tag{4}$$

where $\tilde{g}$ counts effective particle degrees of freedom and greybody factors. Integrating gives the evaporation time

$$t_{\text{evap}} \approx \frac{M_0^3}{3\alpha} \sim \frac{5120\pi G^2 M_0^3}{\hbar c^4} \tag{5}$$

(for $\tilde{g} \sim 1$). Numerically:

$$t_{\text{evap}}(M_\odot) \sim 2 \times 10^{67} \text{ years}, \tag{6}$$

showing that ordinary astrophysical black holes do not evaporate significantly on current cosmic timescales.

1.3 Entropy flow and the Page curve

Define the entropy of radiation emitted up to time t as $S_{\text{rad}}(t)$. The generalized second law requires

$$\frac{d}{dt}\left(S_{BH} + S_{\text{rad}}\right) \geq 0. \tag{7}$$

The Page curve for a unitary evaporation process predicts that the radiation entropy first rises until the Page time t_P, after which correlations reduce the entropy and information is recovered in the radiation. This modern viewpoint constrains any baby-universe information transfer mechanism: information cannot simply be destroyed in a way that violates unitarity unless an explicit non-unitary mechanism is proposed.

2 Information density, interior volume, and the Anas Limit

2.1 Information density concept

We define an *information density* (entropy density) inside the black hole by

$$\rho_{\text{info}}(t) \equiv \frac{S_{\text{int}}(t)}{V_{\text{int}}(t)}, \tag{8}$$

where S_{int} is the entropy effectively contained in the interior and V_{int} is an appropriate interior volume. The horizon area bounds $S_{\text{int}} \leq S_{BH}$, but semiclassical considerations permit considering how S_{int} and V_{int} evolve during late stages.

2.2 Interior volume estimates

For a heuristic estimate of V_{int}, one useful gauge-invariant notion is the Christodoulou–Rovelli maximal slice interior volume $V_{CR}(t) \sim 3\sqrt{3}\,\pi M^2 t$ (in units where $G = c = 1$); details depend on slicing choice and are provided in referenced literature. The essential point is that as M decreases (due to evaporation), the interior volume available to store information can shrink, increasing ρ_{info}.

2.3 Critical information density and the Anas Limit

We postulate that there exists a critical information density ρ_{crit} determined by quantum-gravity physics, above which the semiclassical description breaks down and new dynamical phenomena (a bounce) are triggered. Dimensionally, a natural scale is Planckian density

$$\rho_P \sim \frac{c^5}{\hbar G^2} \sim \frac{m_P c^2}{L_P^3}\,, \tag{9}$$

but other effective critical densities (depending on EoS, particle physics, or exotic phases) may be appropriate. Using a mean-radius criterion, equate the mean radius of a mass M at density ρ to the Schwarzschild radius:

$$\left(\frac{3M}{4\pi\rho}\right)^{1/3} \simeq \frac{2GM}{c^2}. \tag{10}$$

Solving for M yields a characteristic mass scale (the Anas Limit),

$$M_{\text{Anas}} = \gamma \frac{c^3}{G^{3/2}\sqrt{\rho_{\text{crit}}}}, \quad \gamma \equiv \sqrt{\frac{3}{32\pi}} \approx 0.1727470747. \tag{11}$$

Example numerics:

- For $\rho_{\text{crit}} = 10^{18}\ \text{kg}\,\text{m}^{-3}$, $M_{\text{Anas}} \approx 4.3\, M_\odot$.
- For $\rho_{\text{crit}} = 10^{17}\ \text{kg}\,\text{m}^{-3}$, $M_{\text{Anas}} \approx 13.6\, M_\odot$.

These orders of magnitude match compact-star densities and suggest stellar-mass objects are plausible candidates for reaching ρ_{crit} under exotic dynamics.

3 A toy dynamical model: TOV with an effective quantum-pressure term

3.1 Classical TOV system review

The Tolman–Oppenheimer–Volkoff equations for a static spherically symmetric star (metric signature $-+++$) are:

$$\frac{dm}{dr} = 4\pi r^2 \rho(r), \tag{12}$$

$$\frac{dP}{dr} = -\frac{G(\rho + P/c^2)\big(m + 4\pi r^3 P/c^2\big)}{r^2(1 - 2Gm/(rc^2))}. \tag{13}$$

Standard equations of state (EoS) $P(\rho)$ determine equilibrium sequences.

3.2 Effective quantum pressure ansatz

We introduce an effective quantum-pressure correction $P_Q(\rho)$ representing repulsive quantum-gravitation effects (similar in spirit to loop quantum cosmology effective pressures, or gradient-energy dominated terms):

$$P_{\text{eff}}(\rho) = P_{\text{classical}}(\rho) + P_Q(\rho), \tag{14}$$

with a toy parametrization

$$P_Q(\rho) = \alpha \rho^\nu c^2, \qquad \alpha > 0,\ \nu > 1. \tag{15}$$

This extra term is negligible at ordinary densities but becomes dominant near $\rho \sim \rho_{\text{crit}}$.

3.3 Bounce criterion

A qualitative bounce (local halt of collapse) requires the effective central pressure provide sufficient outward force to reverse inward acceleration. An order-of-magnitude criterion is:

$$P_{\text{eff}}(0) \gtrsim \frac{GM^2}{R^4}, \tag{16}$$

or more systematically by integrating the modified TOV numerically one can find parameter regions of (α, ν, M) that permit a rebound. We outline the numerical procedure in the appendix and give representative parameter scans (toy results).

3.4 Toy numerical examples

Using units $G = c = 1$ for code, choose a polytropic classical EoS $P_{\text{classical}} = K\rho^\Gamma$ with $\Gamma = 2$ and K tuned to yield compactnesses typical of neutron-star models. Set $P_Q = \alpha\rho^2$. We find:

- For $M \lesssim 1.4M_\odot$ and $\alpha \lesssim 10^{-4}$, collapse proceeds to singularity (classical behavior).
- For $M \sim 5M_\odot$ and $\alpha \sim 10^{-2}$, central pressure rises steeply near $\rho \sim 10^{18}\,\text{kg/m}^3$ and a bounce occurs in finite simulation time (toy model).

These are illustrative: realistic modeling requires relativistic numerical simulations and a physically motivated P_Q derived from an underlying quantum gravity theory.

4 Entropy accounting, Page-curve compatibility, and information flow

4.1 Entropy budget

Define the total entropy at time t

$$S_{\text{tot}}(t) = S_{BH}(t) + S_{\text{rad}}(t) + S_{\text{other}}(t), \tag{17}$$

which the generalized second law demands is non-decreasing under semiclassical evolution. Any model that transfers interior information into a baby universe must be reconciled with the Page curve: either the transfer occurs in a unitary fashion (i.e., information is encoded in correlations of the radiation or in a larger Hilbert space including the baby universe), or the model departs from unitarity in a controlled way.

4.2 Information transfer scenarios

We outline three conceptual possibilities:

(A) **Unitary transfer via tunneling/nucleation**: the formation of the baby universe is a unitary process when the full parent+child Hilbert space is considered; correlations appear in subtle ways in outgoing radiation (consistent with refined Page curve calculations).

(B) **Unitary re-encoding in a throat/fuzzball**: interior microstates reorganize into a new topological sector (e.g. fuzzball or microstate geometries).

(C) **Non-unitary branch**: effective non-unitary branching occurs where the parent loses accessible degrees of freedom (this requires accepting information non-conservation or embedding in a larger multiverse where global unitarity is preserved but local observers see information loss).

Our preferred approach is (A) or (B): develop the process so that global unitarity is respected. Section 8 discusses observational implications distinguishing these scenarios.

5 Geometry of throat formation: junction conditions and matching

5.1 Setup

Consider matching an interior time-dependent collapse geometry (spherically symmetric) to an exterior geometry which includes an expanding FRW patch connected by a throat. Israel junction conditions determine stress-energy on the matching surface.

5.2 Israel junction conditions

The discontinuity in extrinsic curvature K_{ab} across a timelike shell Σ is related to the shell stress-energy tensor S_{ab} by

$$[K_{ab}]_{-}^{+} = -8\pi G \left(S_{ab} - \frac{1}{2} h_{ab} S \right), \tag{18}$$

where h_{ab} is the induced metric on Σ and $S = h^{ab} S_{ab}$. We set up coordinates on both sides and solve for shell dynamics.

5.3 One-way throat and causal structure

The child universe can be connected by a one-way Einstein–Rosen throat: from the parent's perspective matter/field excitations fall in and are lost behind the throat, while from the child's perspective initial data emerges from a small smooth surface. The matching yields throat radius estimates:

$$r_{\rm th} \approx r_s(1+\epsilon), \quad \epsilon \ll 1, \tag{19}$$

in toy solutions.

6 Baby-universe FRW initial conditions and evolution

6.1 Initial data

We propose the baby universe emerges with an initial scale factor a_b and energy density $\rho_b \sim \rho_{\rm crit}$ determined by the saturated interior. Its Friedmann equation is

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{k}{a^2}. \tag{20}$$

The high initial ρ_b and possible effective vacuum energy from quantum-gravity effects can drive rapid inflationary-like expansion.

6.2 Entropy generation inside the baby universe

If the baby universe begins with ρ_b and negligible entropy per comoving volume (but high total information content), thermalization processes and particle production during a post-bounce phase can generate large entropy, seeding standard cosmological evolution.

6.3 Observables accessible to parent observers

By construction the baby universe is causally disconnected from the parent exterior (except possibly via gravitational-wave signals or transient throat echoes). We discuss potential indirect observational imprints in Section 8.

7 Decoherence, disturbance, and the emergence of classicality

7.1 Decoherence formalism review

In decoherence theory, a system's reduced density matrix loses phase coherence due to entanglement with an environment. For a system initially in superposition $|\psi\rangle = c_1|\phi_1\rangle + c_2|\phi_2\rangle$, interactions with environment produce decoherence with rate $\Gamma_{\rm dec}$ determined by coupling strength, environmental density, and scattering cross-sections.

7.2 Visibility and disturbance operator

We formalize an operator $\hat{D}$ representing perturbations in the "information field":

$$\hat{D} \equiv \int d^3x\, w(x)\, \delta\hat{\rho}(x), \tag{21}$$

with window function $w(x)$ selecting a region. We propose that classical perception corresponds to $\langle \hat{D}^2 \rangle$ exceeding a threshold D_{crit}; equivalently, sufficient environment-induced decoherence renders interference terms negligible for practical observation.

7.3 Marker-on-board toy calculation

Consider a pigment particle of size $a \sim 10^{-6}$ m illuminated by thermal photons at temperature T. Decoherence rate due to photon scattering scales as

$$\Gamma_{\text{ph}} \sim \frac{16\pi^3}{3} \left(\frac{a}{\lambda}\right)^4 c n_\gamma, \tag{22}$$

with photon number density $n_\gamma \propto T^3$ and typical wavelength $\lambda \sim hc/(k_B T)$. At room temperature Γ_{ph} is enormous; even at very low T residual couplings (vacuum fluctuations, blackbody background) provide decoherence. Thus the statement "only at exact 0 K things remain in superposition" is too strong — environmental coupling is the real criterion.

8 Predictions, tests, and observational constraints

8.1 Modified late-stage Hawking emission

If baby-universe formation interrupts semiclassical Hawking evaporation, we might observe deviations from predicted late-stage radiation spectra (high-energy bursts, altered time profiles). However astrophysical BH evaporation is negligible for stellar BHs, so laboratory or primordial BHs are more promising targets.

8.2 Gravitational-wave echoes

Transient throat formation and bounce dynamics can emit characteristic gravitational-wave signals (echoes or bursts). LIGO/Virgo/KAGRA searches for echoes exist; predictions require concrete waveform modeling from full numerical relativity.

8.3 Cosmological acceleration connection

If information leakage into baby universes reduces accessible degrees of freedom in the parent, an effective backreaction may mimic a small cosmological-constant-like term. We present a toy relation:

$$\Lambda_{\text{eff}} \sim \eta \frac{\dot{\mathcal{N}}}{V_{\text{H}}}, \tag{23}$$

where $\dot{\mathcal{N}}$ is the baby-universe formation rate per Hubble volume V_H and η a dimensionless coupling encoding how information loss translates to vacuum-energy-like effects. This is speculative; we include it to indicate how observable correlations might be constructed.

8.4 Constraints from information conservation

If global unitarity holds, any information transferred to a baby universe must be encoded in correlations elsewhere (radiation or other channels). Constraints from Page-curve-compatible models limit how much information can be sequestered without leaving signatures.

9 Mathematical appendices and derivations

Appendix A: Derivation of the Anas Limit

Start with

$$\left(\frac{3M}{4\pi\rho}\right)^{1/3} = \frac{2GM}{c^2}.$$

Cubing both sides:

$$\frac{3M}{4\pi\rho} = \frac{8G^3M^3}{c^6} \quad \Rightarrow \quad 1 = \frac{32\pi G^3 M^2 \rho}{3c^6}.$$

Rearrange:

$$M^2 = \frac{3c^6}{32\pi G^3 \rho} \quad \Rightarrow \quad M = \sqrt{\frac{3}{32\pi}}\frac{c^3}{G^{3/2}\sqrt{\rho}}.$$

Thus M_{Anas} follows with $\gamma = \sqrt{3/(32\pi)}$.

Appendix B: Modified TOV integration algorithm

We outline a stable shooting method: choose central density ρ_c, compute central pressure P_c via EoS, integrate Eqs. (12)–(13) with $P \to P_{\text{eff}}$, stop when P reaches zero at $r = R$, record $M = m(R)$. Sweep parameters (α, ν) and ρ_c to map bounce regions.

Appendix C: Israel junction condition details

Given interior metric $g^-_{\mu\nu}$ and exterior metric $g^+_{\mu\nu}$, compute extrinsic curvatures $K^\pm_{ab}$ on the matching surface and use jump conditions to determine S_{ab}. For spherically symmetric shells, the dynamics reduce to a single equation of motion for shell radius $R(\tau)$:

$$\dot{R}^2 + V(R) = 0,$$

with $V(R)$ an effective potential depending on interior and exterior mass-energy.

10 Discussion: Strengths, weaknesses, and required future work

10.1 Strengths

- Combines well-established physics (BH thermodynamics, decoherence) with speculative but structured quantum-gravity motivated corrections.
- Produces clear, testable channels: GW echoes, modified Hawking spectra, cosmological correlations.
- Introduces a crisp mass–density scale (Anas Limit) enabling parameter surveys.

10.2 Weaknesses and open problems

- The mechanism for converting information saturation into a real bounce requires a concrete microphysical model: LQG effective dynamics, fuzzball microstates, or string-theory tunneling models are candidates.
- Timescale issues: Hawking evaporation is extremely slow for astrophysical black holes; alternative triggers (accretion-driven compression, phase transitions in EoS) must be considered.
- Unitarity and Page-curve consistency must be addressed in explicit models; otherwise the proposal faces severe conceptual challenges.
- Numerical-relativity calculations including quantum corrections are needed to move beyond toy examples.

11 Conclusion

We developed a self-consistent working model tying black-hole information dynamics to baby-universe genesis via an information-saturation criterion and an effective quantum-pressure modified TOV system. The model is intentionally modular: one can replace the toy P_Q with more sophisticated quantum-gravity derived pressures, or couple to specific microstate proposals to examine unitarity. The next stage is quantitative numerical-relativity plus quantum-gravity effective equations to obtain robust waveforms, energy signatures, and formation rates.

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